



1. Compute $\frac{N(E)(W)}{Y(E)(A)(R)}$ when $N = 14, W = 10, Y = 5, A = R = 2, E \neq 0$.

- (A) 1 (B) 2 (C) 4 (D) 7 (E) 14

2. How many days are in the span January 1st, 2010 to December 31st, 2016, including those two dates?

- (A) 1825 (B) 2190 (C) 2192 (D) 2555 (E) 2557

3. Walter's having a New Year's Party. The average age of the 10 kids there was 11. When two new kids of the same age joined the party, the new average age of the kids was 12. How old was one of the kids that just joined?

- (A) 5 (B) 8 (C) 11 (D) 14 (E) 17

4. How many distinct ways can the letters in NEWYEAR be arranged if letters that are the same are indistinguishable?

- (A) 720 (B) 2520 (C) 3780 (D) 5040 (E) 8090

5. Equilateral triangle NEW has equal area as square $YEAR$. What is the ratio NE/AR ?

- (A) $\frac{4}{3}$ (B) $\frac{2\sqrt[4]{27}}{3}$ (C) $\frac{4\sqrt{3}}{3}$ (D) $\frac{3\sqrt{3}}{2}$ (E) 2

6. The digits 2, 0, 1, and 6 are arranged to form a 4-digit number. What is the probability that this number is greater than 2000 and is divisible by 4?

- (A) $\frac{1}{12}$ (B) $\frac{1}{9}$ (C) $\frac{1}{8}$ (D) $\frac{1}{6}$ (E) $\frac{2}{9}$

7. Albert places 3 congruent regular dodecagons together such that they each share 1 side length with the two others, to make a big figure. If the side length of each regular dodecagon is 7, find the area of the gap in the center of large figure.

- (A) $\frac{7\sqrt{13}}{43}$ (B) $\frac{49\sqrt{3}}{4}$ (C) $\frac{49(4-\sqrt{3})}{4}$ (D) $\frac{49\sqrt{3}}{2}$ (E) 49

8. A kindergarten class is taken to a New Year's Magic Performance for 4 days. The magician knows 5 tricks. Each day, he must perform 1 trick for the class. However, he cannot perform the same trick two days in a row, or else the kindergarteners begin to cry. In how many ways can he perform tricks for the 4 days?

- (A) 64 (B) 256 (C) 320 (D) 625 (E) 729

9. A company is receiving applications to perform on New Year's Eve. The waitlist consisted of only singers and rappers, where at the start, $\frac{3}{5}$ of the total were singers. After clearing out 4 rappers and 11 singers, the amount of singers left on the waitlist was equal to $\frac{1}{5}$ the total amount. How many rappers were there on the original waitlist?

- (A) 6 (B) 8 (C) 9 (D) 12 (E) 15

10. Arlmont's football team invented a new sport. The ways to score are as follows: You receive 6 points for a sidekick, and 17 points for a back-kick. David and his friends played one game, and David's team scored 120 points using only sidekicks and back-kicks. How many total kicks did his team score on?

- (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

11. E is the midpoint of YZ on $\triangle XYZ$. F is the midpoint of XZ . A line consisting of point G on EF connects with Z such that $\triangle FGZ$ is $\frac{1}{12}$ of $\triangle XYZ$'s area. If $GE = 7$, then find EF .

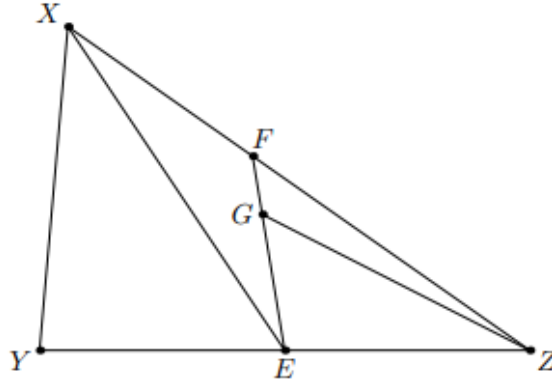


Figure 1: Problem 11

- (A) 8 (B) 10.5 (C) 14 (D) 16.5 (E) 21

12. If for all $x \neq -1$, $\frac{x^2+ax+c}{x+1} = x + b$, where a, b, c are positive, what is c in terms of a ?

- (A) $-a$ (B) $a - 1$ (C) a (D) $a + 1$ (E) 1

13. 5 distinct integers are selected from the first 2016 positive integers such that they can be arranged to form an increasing arithmetic sequence. Let P be the product of the 5 integers. What is the largest integer that must divide P ?

- (A) 1 (B) 3 (C) 5 (D) 15 (E) 30

14. Robert plays with some number of red and yellow blocks in which he wants to make towers out of. Each smart-tower consists of 3 red blocks and a number of yellow blocks. Robert first makes a design of smart-towers using 4 yellow blocks per tower, using up all his yellow blocks, leaving 6 red blocks. He now restarts and makes smart-towers out of 2 yellow blocks. He uses up all his red blocks, leaving 6 yellow blocks. How many blocks total does Robert have?

- (A) 37 (B) 41 (C) 42 (D) 48 (E) 52

15. A college has received 30 applications from students. For each student, it has the choice to accept, reject, or defer them, regardless of admission result for the other students. Let S be the number of possible outcomes of the admission process for all 30 students combined. Compute the remainder when S is divided by 12.

(A) 1 (B) 3 (C) 5 (D) 9 (E) 11

16. The lines $y = x + 1$ and $y = 1$ intersect at $A = (0, 1)$. Let point B be $(3\sqrt{2}, 1 + 3\sqrt{2})$. $ABCD$ is a rhombus, with C above the x -axis and D in the form $(n, 1)$ such that n is an integer. What is the sum of the possible values of n ?

(A) -6 (B) -4 (C) -2 (D) 0 (E) 6

17. Let F be the center of a circle with radius 6 and diameter AB . C is a point on the circle such that minor arc AC has length 4π . What is the area of $\triangle ABC$?

(A) $6\sqrt{2}$ (B) $6\sqrt{3}$ (C) $9\sqrt{2}$ (D) $9\sqrt{3}$ (E) $18\sqrt{3}$

18. In preparation for his Spanish test, Leon has made 8 notecards for 4 terms. Each term has 2 notecards corresponding to it—one of the term in English, and one of the term in Spanish. If Leon randomly selects 4 notecards, what is the probability that exactly one pair of cards correspond to the same term?

(A) $\frac{7}{16}$ (B) $\frac{9}{16}$ (C) $\frac{47}{70}$ (D) $\frac{24}{35}$ (E) $\frac{4}{5}$

19. $3f(x) + 2f\left(\frac{\sqrt{6}}{6x}\right) = x^2$, for all x except 0. What is $12f(2)$?

(A) $\frac{127}{10}$ (B) $\frac{143}{10}$ (C) 15 (D) $\frac{127}{5}$ (E) $\frac{143}{5}$

20. In regular hexagon $ABCDEF$ with side length 6, let M be the center, and let points A', B', C', D', E', F' be on AM, BM, CM, DM, EM, FM respectively such that $AM : A'M = BM : B'M = CM : C'M = DM : D'M = EM : E'M = FM : F'M = \frac{3}{2}$. Now let $A'', B'', C'', D'', E'', F''$ be the midpoints of AB, BC, CD, DE, EF, FA respectively. What is the perimeter of concave dodecagon $A'A''B'B''C'C''D'D''E'E''F'F''$?

(A) $24\sqrt{3}$ (B) $\frac{36\sqrt{17}}{5}$ (C) $12\sqrt{7}$ (D) 36 (E) 48

21. t_n is generated by combining consecutive positive integers starting from 1 to n into an integer, the end result. For example, t_{9000} can be expressed as

123456789101112...2014201520162017...89999000. Integer T is formed by taking the first 2016 digits of t_{2016} . What is the remainder when T is divided by 9?

(A) 0 (B) 1 (C) 3 (D) 4 (E) 8

22. Jason has 5 indistinguishable white pawns and 4 indistinguishable black pawns that he wishes to place on a 3×3 chessboard (where each square of the chessboard is distinct) such that there is exactly one pawn per square. He describes a position as drawish if no row or column has pawns of all the same color. How many possible setups are drawish?

(A) 36 (B) 45 (C) 69 (D) 75 (E) 93

23. $\triangle ABC$ has $BC = 6$. The internal angle bisector of $\angle BAC$ meets BC at E with $AE = 9$. Let I be the intersection of the internal angle bisectors of $\triangle ABC$. AE is extended past E to D such that $BD \parallel IC$ and $CD \parallel IB$. What is $\frac{AI}{DI}$?

(A) $\sqrt{3} - 1$ (B) $\frac{\sqrt{10}}{2}$ (C) $2\sqrt{3}$ (D) 3 (E) $\sqrt{10}$

24. Let a, b, c be the roots of the polynomial $x^3 - 6x^2 - 37x + 90$. Given that $a^3 + b^3 + c^3 = 612$, what is $\frac{(a^2 - ab - ac)(b^2 - ab - bc)(c^2 - ac - bc)}{abc}$?

(A) 384 (B) 744 (C) 800 (D) 890 (E) 1000

25. Points A, B, C, D are on sphere Z such that AB bisects CD . If segments $AB = 15$, $CD = 12$ and $AC = 5$, find the area of $ACBD$.

(A) $20\sqrt{14}$ (B) $54\sqrt{7}$ (C) $12\sqrt{42}$ (D) $\frac{104\sqrt{14}}{5}$ (E) $24\sqrt{14}$

Time limit: 75 minutes